

ABHINAV A

AI/ML Engineer

Kannur, India • +91 9961591699 • abhinava818@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

MCA graduate (2026) specializing in AI and machine learning, with hands-on ownership of ML solutions from data analysis and model training to API deployment. Built a federated credit-default system (0.776 ROC-AUC, 85.3% recall) with retraining and model-version tracking, an embedding-based semantic search app using prompt-based LLM workflows (Gemini API), and a real-time computer vision application. Trained and guided participants in AI, ML, and data science at Inker Robotic Solutions. Completed the Machine Learning Specialization (Stanford/DeepLearning.AI) and the Deep Learning Specialization (DeepLearning.AI).

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript

Generative AI & LLMs: LLM application development, prompt engineering, retrieval-grounded generation (RAG-style), semantic search, embeddings (SentenceTransformers), Gemini API, structured information extraction, unstructured data processing, output validation

Machine Learning: Applied Machine Learning, Model Training, Model Evaluation (ROC-AUC, precision, recall, F1), Model Optimization, Scikit-learn, XGBoost, Random Forest, Logistic Regression, ensemble learning (stacking), feature engineering, class-imbalance handling, cross-validation, threshold optimization, federated learning (FedAvg)

Deep Learning: TensorFlow, Keras, neural networks, MediaPipe pose estimation

MLOps & Deployment: MLOps (Machine Learning Operations), Application Deployment, Continuous Deployment, Continuous Monitoring, System Monitoring, model serving (joblib), model retraining workflows, model-version tracking, FastAPI, REST APIs, Docker, Git, Linux

Data Engineering & Quality: Data Collection, Data Preprocessing, Data Validation, Data Quality, Data Integrity, Data Maintenance, PostgreSQL, SQLAlchemy, NumPy, Pandas

Coursework: Transformers, RNN/LSTM, CNNs, recommender systems, clustering, anomaly detection, reinforcement learning

PROFESSIONAL EXPERIENCE

AI, ML & DATA SCIENCE TRAINER

Inker Skill Fest Program | Inker Robotic Solutions Pvt. Ltd. | May 2026

- Delivered and guided hands-on training for program participants in Artificial Intelligence, Machine Learning, Data Science & Analytics, and Python with its libraries, using practical projects.
- Received a written appreciation from the CEO for in-depth sessions and a practical teaching approach that contributed to the program's successful delivery.

FULL STACK DEVELOPER INTERN

Inker Robotic Solutions Pvt. Ltd. (Inker Intelligence division) | Feb 2026 – May 2026

- Designed and built a multi-tenant WhatsApp-based ticket booking and show management SaaS platform end to end: requirements and BRD/SRS documentation, system architecture, frontend and backend development, database design, WhatsApp workflows, QR ticket generation, payment integration, ticket validation, and testing.

- Independently designed a 10-day Full Stack Development course for internship trainees and conducted a 5-day AI and Machine Learning training program.

TECHNICAL PROJECTS

Federated Learning–Based Credit Card Default Prediction

Python, Scikit-learn, XGBoost, TensorFlow/Keras, FastAPI, PostgreSQL, SQLAlchemy, JWT

- Built a federated learning system for credit card default prediction across 3 simulated bank branches (UCI dataset, 30,000 records): engineered 5 features that ranked as top predictors, trained four model types per branch with class-imbalance handling, and aggregated them with FedAvg, tree merging, and stacking.
- Fixed neural-network FedAvg instability with fine-tuning on a small shared subset (10% of data) and threshold calibration. The global ensemble reached 0.776 ROC-AUC on a 4,500-record holdout never used in training, with 85.3% recall at 31.5% precision at a 0.30 threshold tuned to catch defaulters.
- Built a role-based FastAPI and PostgreSQL backend with JWT authentication, audit logging, and a retraining workflow.

Syllabus Smart Search

Django, SentenceTransformers, Gemini API, SQLite, ReportLab

- Built a retrieval-grounded semantic search app using a prompt-based LLM workflow: Gemini generates a hidden reference definition for a syllabus topic, which is embedded with SentenceTransformers and matched against pre-vectorized textbook paragraphs, so results shown to the user come only from the uploaded textbook.
- Implemented PDF text extraction, same-page diagram retrieval, confidence scoring, and PDF export of textbook extracts plus Gemini-generated study notes, with authentication and per-user textbook isolation.

Invoice-PO Matcher

Node.js, Express, React, Vite, Gemini-2.5-Flash-Lite

- Built a document-comparison tool that uses Gemini-2.5-Flash-Lite to extract vendor, totals, and line items from invoices and purchase orders (with an OCR fallback), scores how closely the two match, and shows the results in a React table with CSV export.

AI Form Corrector

Next.js, TypeScript, MediaPipe Pose Landmarker, FastAPI, Docker, Recharts

- Built a real-time workout form analysis app using in-browser MediaPipe pose landmarks to compute joint angles, count reps with a finite-state machine with hysteresis, grade squat depth, detect torso lean, and give spoken coaching feedback.
- Designed for on-device processing so video never leaves the browser, with local session history and progress charts for volume and form quality.

EDUCATION

Master of Computer Applications (MCA) CGPA: 8.2/10	2024 – 2026
College of Engineering Thalassery, Kannur Cochin University of Science and Technology	
Bachelor of Computer Applications (BCA) CGPA: 7.6/10	2021 – 2024
SNG College of Advanced Studies, Kannur Kannur University	

CERTIFICATIONS & COURSES

- Machine Learning Specialization | DeepLearning.AI & Stanford | Coursera | 2026
- Deep Learning Specialization | DeepLearning.AI | Coursera | 2026
- Machine Learning in Production | DeepLearning.AI | Coursera | 2026
- Python DSA + LeetCode | Udemy | 2026
- Full-Stack Web Development | Udemy | 2025